

Implementation Plan — Spatial Degradation of Hepatocyte Zonation

Exact inputs, algorithms, formulas, and the code file for every step

Companion to the stubs in `src/steps/` and the reference implementation `src/pipeline.py`

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This document expands each stub in `src/steps/` into exact inputs, algorithms, formulas, data shapes, outputs, and acceptance checks. It is the “fill-in-the-blanks” companion: the docstrings say *what*; this says *exactly how*. Reference implementations for the statistical steps already live in `src/pipeline.py` (the integrated donor-level script), and each step section below ends with a **Code:** line naming the file(s) that implement it.

1 Conventions used throughout

1.1 Paths and the config module

All paths are imported from `src/config.py`; no hard-coded paths appear in the steps.

- **PAPER1** — directory of the Paper 1 (disease atlas) inputs: `counts.mtx`, `genes.txt`, `barcodes.txt`, `cell_metadata.csv`, `metadata_all_cells.csv`.
- **PAPER2_TRAIN** — `paper2_train.npz`, the classifier training bundle.
- **SIGNATURES** — the `signatures/` directory of zonation gene lists.
- **FIGURES, TABLES** — output directories for artefact figures and CSVs.
- `config.signature_files(name)` — returns the (`pericentral`, `periportal`) file pair for a chosen signature set (see Section 1.5); `config.DEFAULT.SET` selects the default.

1.2 Matrix orientation, donor, and stage

- **Matrix.** `counts.mtx` is $genes \times cells$ ($30,117 \times 69,426$), raw integer UMIs. Row $i \leftrightarrow genes.txt[i]$; column $j \leftrightarrow barcodes.txt[j]$.
- **Donor** = `metadata_all_cells.csv["Patient.ID"]` (about 47 donors).

- `Stage = cell.metadata.csv["stage"]`, an ordered factor mapped to ranks $0 \dots 4$ by the helper S2R:

Healthy control < NAFLD < NASH w/o cirrhosis < NASH with cirrhosis < end stage $\mapsto 0, 1, 2, 3, 4$.

1.3 The golden rule — donor is the unit of inference

The unit of inference is the **donor**, not the cell. For every hypothesis we aggregate cells into *one number per donor*, then run the statistical test on the ~ 47 donor values. Cell-level p-values are pseudoreplication and are never reported as the primary result.

1.4 The A4 table (the shared backbone)

Step 4 emits a per-cell table consumed by Steps 5–8 with columns

```
cell_id, donor, stage, zonation.coord, pc, pp, plasticity,
zone.p0, zone.p1, zone.p2, entropy.
```

1.5 Signature sets — transcriptome-wide by default

The zonation coordinate is read from a **gene program**, and the size of that program is a first-class design choice. The motivation is to read position from the *whole* zonation program, not from a handful of marker genes.

Primary set: full (transcriptome-wide). The default signature is the full transcriptome-wide zonation program from Paper 2:

- `signatures/pericentral.full.txt` — 1,273 pericentral genes,
- `signatures/periportal.full.txt` — 364 periportal genes,

for a total of $\approx 1,637$ *significantly-zonated* hepatocyte genes (Paper 2, $q < 0.05$, well-expressed, and present in Paper 1). This is `config.DEFAULT_SET = "full"`. **Both** the coordinate *and* the classifier use this full set. The classifier’s `paper2.train.npz` cache now stores the *union of all tiers* ($\sim 1,700$ genes), so any set can be sliced from it by gene name; re-run `src/prep/02.convert-paper2.mat.py` once if the cache predates this change.

Robustness / baseline alternatives. Smaller sets are kept as robustness and baseline checks, each selectable via `config.signature_files(name)`:

Set name	Size (PC / PP)	Role
full	1273/364	primary, transcriptome-wide (default)
expanded	$\sim 102/91$	ranked top genes, ~ 100 each
core	13/8	curated canonical markers
paper2.landmark	13/8 exact	Paper 2 landmark baseline

The decision: full is the result, paper2.landmark is the audit. We report both, but they do *different jobs* — this is not a 50/50 hedge. `full` is primary because (i) it *is* the contribution (read position from the entire program, not markers by eye); (ii) it is the *more robust* ruler in disease — the coordinate is a *mean* of z-scored genes, so disease-corruption of any single marker is diluted (e.g. CYP2E1, canonical pericentral, is induced in NASH for non-spatial reasons and could swing a 13-gene ruler, but is 1-of-~1,600 here); and (iii) only a large set makes the H2 “which programs collapse” question and its non-circular held-out-gene test (Step 7) meaningful. `paper2.landmark` is the *credibility control*: a canonical, Paper-2-faithful, fully interpretable 13+8 ruler. If it shows the *same* collapse, the signal is not an artefact of gene-set size; if the two *disagree*, that is itself a finding (a warning the ruler may be disease-sensitive). **Per hypothesis:** H1 (collapse) + ruler/classifier $\rightarrow$ `full` primary, `paper2.landmark` control; H2 $\rightarrow$ `full` with the held-out repeated split; H3 $\rightarrow$ de-zonation score from `full`. `expanded/core` are an optional robustness gradient. **Why the original plan said “landmark”:** marker-based reconstruction is the field-standard, validated method (Paper 2; Halpern et al. 2017) — the conservative starting point; `full` is the deliberate upgrade.

The PC/PP imbalance (1,273 vs 364) does not bias the coordinate. This is a *feature-set* asymmetry, not ML class imbalance (which concerns label counts, e.g. the classifier’s zone terciles). The coordinate is $\text{mean}_z(\text{PC}) - \text{mean}_z(\text{PP})$; because each arm is a *mean* (not a sum), the 3.5:1 count ratio carries *no* magnitude bias. It does create a *precision* asymmetry — the PP arm averages fewer genes, so it is noisier and the periportal end is estimated less reliably. Mitigations: (1) standardise each arm to unit variance across cells *before* subtracting (equalises the arms regardless of count); (2) a size-matched run — top-364 strongest-zonated PC genes vs the 364 PP — as a robustness check; (3) optionally strength-weight genes by Paper 2 zonation effect size, which shrinks the imbalance by down-weighting the long tail of weak PC genes. The imbalance itself is part biology (zonation skews pericentral here) and part the selection thresholds ($q < 0.05 + \text{p90 expression} + \text{center-of-mass split}$).

Code: `src/config.py` (PAPER1, PAPER2_TRAIN, SIGNATURES, FIGURES, TABLES, `config.DEFAULT_SET`, `config.signature_files("full")`).

2 Step 1 — Build the ruler (prep)

2.1 Inputs

- Paper 2 landmark CSVs $\rightarrow$ `signatures/*_paper2.landmark.txt` (the 20+20 baseline).
- `supplementary_table.8.xlsx`, sheet Hepatocyte $\rightarrow$ `signatures/*_expanded.txt` (ranked, ~ 100 each), and the full $q < 0.05$ program $\rightarrow$ `signatures/*_full.txt`.
- `combined_scrnaseq_atlas_M5M6M7M8.mat` $\rightarrow$ `paper2_train.npz`.

2.2 Training labels (the classifier’s y)

Each Paper 2 nucleus receives a zone label:

- **Ground truth (target).** Port `parse_scrnaseq_combined_atlas.m`: Paper 2 mapped its *spatial* (Visium) zonation onto its snRNA nuclei; use that zone index, binned to 3 classes.
- **Fallback (current `paper2_train.npz`).** Compute on Paper 2’s own hepatocytes

$$\text{zone_label} = \text{tercile}(\text{score}(\text{PC}) - \text{score}(\text{PP})) \in \{0, 1, 2\}$$

(0 portal, 1 mid, 2 central).

2.3 Output and acceptance

Artefact A1. Acceptance: 3 zone classes present with balanced-ish counts; PC genes rise and PP genes fall along the zone index (profile plot).

Code: `src/02.convert_paper2_mat.py`, `src/03.build_signatures.py`, the `signatures/` directory; profile plot `src/plotting/artefacts.py:plot_a1_reference_profiles`.

3 Step 2 — Load & QC

3.1 Inputs

PAPER1/{`counts.mtx`, `genes.txt`, `barcodes.txt`, `cell_metadata.csv`, `metadata_all_cells.csv`}.

3.2 Algorithm

1. `M = scipy.io.mmread(PAPER1/"counts.mtx").tocsc()` $\rightarrow$ genes $\times$ cells sparse.
2. `genes = [l.strip() ...]`, `bars = [l.strip() ...]` (order = matrix rows / columns).
3. `meta = read_csv(cell_metadata).set_index("cell_id").reindex(bars)` $\rightarrow$ `stage = meta["stage"]`.
4. `allm = read_csv(metadata_all_cells).set_index("cell_id")`; auto-detect the donor column from `["Patient.ID", "patient", "donor", "orig.ident", "sample"]`; `donor = allm[col].reindex(bars)`.
5. `libsize = asarray(M.sum(0)).ravel()` (per-cell total UMIs).
6. **QC sanity only (do *not* re-filter).** Confirm the kept cells are hepatocytes — mean ALB expression is high; print per-stage and per-donor counts. Optional sensitivity: flag the lowest-`nCount_RNA` decile for a robustness re-run, do not delete.

3.3 Normalization (used by scoring; the `zrows()` helper)

For a gene row x , first CP10k then $\log 1p$:

$$v = \log\left(1 + \frac{x}{\text{libsize}} \cdot 10^4\right), \quad (1)$$

$$z = \frac{v - \text{mean}(v)}{\text{std}(v)} \quad (\text{if } \text{std}(v) = 0, z = 0). \quad (2)$$

3.4 Output and acceptance

Output A2: `M`, `genes`, `bars`, `stage`, `donor`, `libsize` plus a per-stage / per-donor count table.

Acceptance: 69,426 hepatocytes; 5 stages with counts 3750/7073/31903/3603/23097; ALB clearly expressed.

Code: `src/steps/step2_load_qc.py` (ref `src/pipeline.py:load`).

4 Step 3 — Harmonize genes

4.1 Inputs

A2 genes plus the signature gene sets (and `paper2_train.feats` for the classifier).

4.2 Algorithm

1. Reconcile symbols (and Ensembl IDs where present) across Paper 1, Paper 2, and the signatures; report the mapping rate $|\text{shared}|/|\text{signature}|$.
2. Restrict signatures and classifier features to the shared set. If any signature drops below ~ 15 genes, widen it via the `expanded` list.
3. Choose batch-robust features: per-gene z-score (above) and/or a rank transform, so the score measures portal-vs-central rather than snRNA-vs-Visium platform.

4.3 Output and acceptance

Output A3: a harmonization report (mapping rate, surviving signature sizes) plus the frozen gene sets. **Acceptance:** $\geq 80\%$ of signature genes survive the intersection.

Code: `src/steps/step3_harmonize.py`.

5 Step 4 — Place every cell

5.1 Step 4a — Signature scoring

For a gene set S , use the **background-subtracted score** (Scanpy `score_genes` style). Let Y be the z-scored log 1p-CP10k expression (genes $\times$ cells). For cell c ,

$$\text{score}_S[c] = \frac{1}{|S|} \sum_{g \in S} Y[g, c] - \frac{1}{|B|} \sum_{g \in B} Y[g, c],$$

where B is a control set drawn to match the expression-bin distribution of S . Subtracting B removes each cell's overall expression / depth baseline, so the score reflects S specifically.

Then:

$$\text{pc} = \text{score}(\text{PERICENTRAL}), \quad \text{pp} = \text{score}(\text{PERIportal}), \quad (3)$$

$$\text{coord} = \text{pc} - \text{pp} \quad (\text{signed; } > 0 \text{ pericentral, } < 0 \text{ periportal}), \quad (4)$$

$$\text{plasticity} = \text{score}(\text{PLASTICITY}) \quad (\text{signatures/plasticity.txt: KRT7, KRT19, SOX9, SOX4, KRT23, NCAM}) \quad (5)$$

The default gene sets are the `full` program (Section 1.5). The simpler `src/pipeline.py:score` uses the mean of z-scored signature genes *without* an explicit control set; the `score_genes` background form above is the upgrade.

Code: `src/steps/step4_score.py` (ref `src/pipeline.py:score`); plot `src/plotting/artefacts.py:plot_a4_coordina`

5.2 Step 4b — Classifier + entropy

1. Train on `paper2_train.npz`: X (nuclei $\times$ feature genes, CP-normalized) and $y = \text{zone_label} \in \{0, 1, 2\}$. With the `full` set the features are the full zoned gene set (*not* the 40 landmarks).
2. `Pipeline(StandardScaler(), LogisticRegression(multi_class="multinomial", C=..., max_iter=...))`. Prefer a **calibrated** logistic model (well-behaved probabilities) over a random forest.
3. **Evaluate on a held-out Paper 2 split FIRST** (`train_test_split`, stratified by donor): accuracy and confusion matrix; must clearly beat chance (1/3).
4. Apply to Paper 1 (same features, same normalization): `P = clf.predict_proba(X1) → zone_p0..2`.
5. **Entropy** per cell:

$$H[c] = - \sum_k p_k \log p_k$$

(k over the 3 zones, natural log, range $[0, \log 3]$). Low H = confident zone; high H = de-zoned / confused.

6. **4c agreement.** `spearman(zonation_coord, expected_zone)` on Paper 1 (especially healthy) should exceed ~ 0.5 .

Output A4: merge `zone_p0..2`, `entropy` into the per-cell table.

Code: `src/steps/step4b_classifier.py` (ref `src/classifier.py`).

6 Step 5 — Validate on healthy (GATE)

6.1 Input and algorithm

Input: A4 restricted to Paper 1 healthy donors. For each validation marker,

$$\rho = \text{spearman}(\text{coord}[h], \text{expr}_{\text{marker}}[h]).$$

- Pericentral CYP2E1, CYP1A2, GLUL, PCK2 $\rightarrow$ expect $\rho > 0$.
- Periportal ASS1, ALDOB, PCK1, HAL $\rightarrow$ expect $\rho < 0$.
- Classifier entropy in healthy should be low (peaked predictions).

6.2 Output and the gate

Output A5: a bar of ρ per marker plus the values. **Acceptance / GATE:** pericentral $\rho > 0$ and periportal $\rho < 0$. *Do not start Step 6 until this passes.* If it fails: widen to `expanded` signatures, use rank features, revisit normalization.

Code: `src/steps/step5_validate.py` (ref `src/pipeline.py:validate`); plot `src/plotting/artefacts.py:plot_a5_va`

7 Step 5b — Is the ruler still a ruler?

Per stage s , on the cells of that stage:

1. **Internal coherence.** Mean pairwise correlation among pericentral genes (and among periportal genes). Healthy = high; if it crashes, the module is dissolving (not just de-zoning).

2. **Cross-program anti-correlation.** $\text{corr}(\text{pc}, \text{pp})$ across cells. Healthy $\approx$ strongly negative; weakening toward 0 = true de-zonation (robust to global shifts).
3. **Split-half reproducibility.** Randomly split each signature in two, build two coordinates, and correlate them. A drop means the signature is no longer self-consistent.
4. **Program-off vs restriction-lost.** Track each module's mean magnitude (is it switched off?) versus the rate of cells co-expressing both ends (is positional restriction lost?).

Output A5b: per-stage curves; tells you *which kind* of breakdown each stage is.

Code: `src/steps/step5b_ruler_diagnostics.py`; plot `src/plotting/artefacts.py:plot_a5b_ruler`.

8 Step 6 — Collapse curve (H1)

8.1 Per-donor metrics

For donors with ≥ 30 cells, one row per donor:

$$\text{spread}[d] = \text{std}(\text{coord}[\text{donor} = d]) \quad (\text{narrows with disease}), \quad (6)$$

$$\text{anticorr}[d] = \text{spearman}(\text{pc}[\text{donor} = d], \text{pp}[\text{donor} = d]) \quad (\text{rises toward 0}), \quad (7)$$

$$\text{entropy}[d] = \text{mean}(H[\text{donor} = d]) \quad (\text{rises, if classifier run}), \quad (8)$$

$$\text{stage_rank}[d] = \text{S2R}[\text{mode}(\text{stage}[\text{donor} = d])]. \quad (9)$$

8.2 Ordered-trend test on the ~ 47 donor values

- **Spearman.** ρ , $p = \text{spearmanr}(\text{stage_rank}, \text{metric})$; with no ties

$$\rho = 1 - \frac{6 \sum_d d^2}{n(n^2 - 1)}.$$

- **Jonckheere–Terpstra** (ordered alternative across the 5 stage groups):

$$J = \sum_{a < b} U(\text{group}_a, \text{group}_b),$$

where U counts pairs with $x_b > x_a$. Standardize with the null mean

$$\mu_J = \frac{N^2 - \sum_s n_s^2}{4}$$

and the standard JT variance σ_J^2 to obtain z and p .

8.3 Confidence and negative control

- **Donor bootstrap CI.** Resample donors with replacement $B = 2000$ times, recompute ρ , and take the 2.5/97.5 percentiles.
- **Negative control (permutation).** Permute `stage_rank` across donors $B = 2000$ times, recompute $|\rho|$, and report

$$p_{\text{perm}} = \frac{\#\{\text{null} \geq \text{observed}\} + 1}{B + 1}.$$

Output A6: `collapse_per_donor.csv` plus a plot (donor points + per-stage mean) and per-metric ρ , 95% CI, p , p_{perm} . **Acceptance:** spread $\downarrow$, anticorr $\rightarrow 0$, (entropy $\uparrow$); CI excludes 0; p_{perm} small.

Code: `src/steps/step6_collapse.py` (ref `src/pipeline.py:collapse`); plot `src/plotting/artefacts.py:plot_a6.co`

9 Step 7 — Zone-resolved DE (H2)

9.1 Bin cells into zones

By coordinate terciles: `terc = quantile(coord, [1/3, 2/3])`, `zone = digitize(coord, terc)  $\rightarrow$  0` portal, 1 mid, 2 central.

9.2 Pseudobulk (the donor-level move)

For each zone $z \in \{\text{portal}, \text{central}\}$, for each donor d with ≥ 20 cells in that zone, build one profile:

$$\text{cells} = (\text{zone} = z) \ \& \ (\text{donor} = d), \quad (10)$$

$$\text{cp} = \text{per-cell CP-normalized counts} = \text{counts}/\text{colsum}, \quad (11)$$

$$\text{PB}[d, :] = \log_1 p \left(\text{mean}_{\text{cells}}(\text{cp}) \cdot 10^4 \right) \quad (\text{donor} \times \text{gene}), \quad (12)$$

$$\text{ranks}[d] = \text{stage_rank}[d]. \quad (13)$$

9.3 Test per gene (detected in $\geq 10\%$ of the zone’s donors)

- **MVP.** `rho, p = spearman(PB[:, g], ranks)` across donors.
- **Cleaner H2 (the real test).** OLS interaction on the pseudobulk

$$\text{expr}_g \sim \text{coord} + \text{stage} + \text{coord}:\text{stage};$$

the `coord:stage` coefficient answers “does this gene’s zonal slope weaken with stage?” — i.e. lost zonation, not just lost level.

9.4 Multiple testing (BH-FDR)

Sort the p -values ascending; the threshold is $p_{(k)} \leq \frac{k}{m}\alpha$; take the largest such k ; everything below it is significant; the q -value is the monotone-adjusted p . Report at $q < 0.05$.

9.5 Circularity guards (mandatory)

The coordinate is built *from* the signature genes, so testing those same genes for “collapse” is circular. Two guards:

- **Signature flag.** Flag `is_signature` genes and report significant counts BOTH including and EXCLUDING them; “driver” claims use non-signature genes.
- **Held-out gene split (the key fix).** Build the coordinate on one random half of the gene set and test collapse on the disjoint other half.

Repeated random held-out splits. Because any single random partition could be lucky, repeat it K times ($K \approx 20\text{--}50$ random splits) and report the *distribution* of the result across splits:

- the **median effect** across splits, and
- the **fraction of splits** that are significant.

A result that holds across most random splits is robust; one that depends on a particular partition is not. Practical notes:

- **Fix the RNG seed** for reproducibility.
- **Optionally stratify** the split so each half keeps both PC and PP genes.
- This is **meaningful only with the large full set**, not 20 landmarks — you cannot split 40 genes into two informative halves.

Output A7: `de_portal.csv`, `de_central.csv` (gene, effect, p, q, is_signature) plus a volcano plot. **Acceptance:** the significant set survives excluding signature genes *and* holds across most random splits.

Code: `src/steps/step7_de.py` (ref `src/pipeline.py:de`); plot `src/plotting/artefacts.py:plot_a7_volcano`.

10 Step 8 — De-zonation $\leftrightarrow$ plasticity (H3)

10.1 De-zonation proxy

Per cell, distance of coord from the committed ends:

$$\text{dez} = - \left| \frac{\text{coord} - \text{median}(\text{coord})}{\text{std}(\text{coord})} \right|$$

(near the middle $\Rightarrow$ more de-zonated); high entropy is an alternative proxy.

10.2 The confound and the within-donor test

Both `dez` and `plasticity` rise with stage, so a *pooled* correlation is a fake-positive. Test **within donor**, then aggregate:

- Per donor d (≥ 30 cells): $r[d] = \text{spearman}(\text{dez}[\text{donor} = d], \text{plasticity}[\text{donor} = d])$; report $\text{mean}(r)$ and the % of donors with $r > 0$.
- **Regression (formal version).** OLS $\text{plasticity} \sim \text{dez} + C(\text{stage}) + C(\text{donor})$; the `dez` coefficient is the association *after* removing stage and donor; report its sign and p .
- **Optional two-group view.** `MannWhitneyU(plasticity[de-zonated], plasticity[zonated])` within stage, with a rank-biserial effect size.

Output A8: a stage-stratified scatter plus the within-donor statistic and the `dez` coefficient. **Acceptance:** the within-donor (not pooled) effect is reported with sign and p , honestly (positive or null).

Code: `src/steps/step8_plasticity.py` (ref `src/pipeline.py:plasticity`); plot `src/plotting/artefacts.py:plot_a8`.

11 Step 9 — Bonus: collapse vs inherited risk

11.1 Inputs

- `hits` = the Step-7 hit list ($q < 0.05$, non-signature).
- `risk` = Paper 3 risk-variant target genes.
- `background` = genes **testable** in Step 7 (detected + tested), ideally expression-matched.

11.2 Test (hypergeometric enrichment)

With $k = |\text{hits} \cap \text{risk}|$, $n = |\text{hits}|$, $K = |\text{background} \cap \text{risk}|$, $N = |\text{background}|$:

$$p = \text{hypergeom.sf}(k - 1, N, K, n) \quad (\text{scipy}), \quad (14)$$

$$\text{fold} = \frac{k/n}{K/N}, \quad (15)$$

with BH applied across the few tests.

Output A9: fold + CI + q . **Acceptance:** the background is well-defined and the result is stated with caveats (associational). Strictly optional.

Code: `src/steps/step9_bonus_enrichment.py`.

12 Plotting machinery

Function	Input	Draws
<code>plot_a1_reference_profiles</code>	gene $\times$ zone means	PC rising / PP falling along zone
<code>plot_a4_coordinate</code>	A4 table	coord histogram + 2 marker scatters
<code>plot_a5_validation</code>	A4 healthy + markers	bar of ρ , colored by expected sign
<code>plot_a5b_ruler</code>	per-stage diagnostics	coherence / anti-corr / split-half curves
<code>plot_a6_collapse</code>	per-donor metrics	donor points + per-stage mean
<code>plot_a7_volcano</code>	DE table	effect vs $-\log_{10} q$ volcano
<code>plot_a8_plasticity</code>	A4 table	stage-stratified dez-vs-plasticity scatter

Code: `src/plotting/artefacts.py`.

13 Drivers

- **Full pipeline driver.** `src/run_all.py` — runs Steps 2–9 end to end and writes all artefacts.
- **Positive control.** `src/run_p2_validation.py` — runs the ruler against Paper 2’s own data, where the answer is known, as a sanity check on scoring and the classifier.

Code: `src/run_all.py`, `src/run_p2_validation.py`.